

$L_{1/2}$ Regularization for Image Denoising

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Problem Statement

- ▶ Goal: Recover clean image x from noisy image $y = Ax + \eta$
- ▶ Classical methods:
 - ▶ L_2 : Smooth but blurs edges
 - ▶ L_1 : Promotes sparsity but introduces bias
- ▶ Idea:
 - ▶ Use $L_{1/2}$ regularization
 - ▶ Better sparsity + better reconstruction

Key Idea

- ▶ Solve sparse coding problem:

$$\min_x \frac{1}{2} \|Dx - y\|^2 + \lambda \|x\|_p$$

- ▶ Compare:
 - ▶ OMP (greedy)
 - ▶ L_1 (convex)
 - ▶ $L_{1/2}$ (non-convex)
- ▶ Expectation:
 - ▶ Better sparsity $\rightarrow$ better denoising

Algorithms Used

- ▶ OMP (Baseline)
 - ▶ Greedy atom selection
 - ▶ Fast but less accurate
- ▶ L_1 (ISTA)
 - ▶ Soft-thresholding
 - ▶ Iterative but slower
- ▶ $L_{1/2}$ (IRLS)
 - ▶ Non-convex formulation
 - ▶ Strong sparsity

IRLS Method

- ▶ Convert $L_{1/2}$ problem into weighted L_2
- ▶ Iterative update:

$$x = (D^T D + \lambda W)^{-1} D^T y$$

- ▶ Weights depend on previous estimate
- ▶ Advantage:
 - ▶ Simple implementation
 - ▶ Efficient in practice

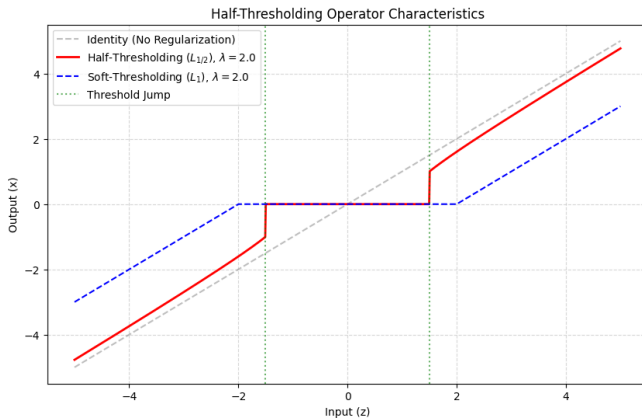
Half-Thresholding Operator

- ▶ Arises from solving:

$$\min_x \frac{1}{2}(x - z)^2 + \lambda|x|^{1/2}$$

- ▶ Compared to:
 - ▶ Soft-thresholding (L_1)
 - ▶ Hard-thresholding (L_0)
- ▶ Key idea:
 - ▶ Less bias than L_1
 - ▶ Stronger sparsity

Operator Visualization



Why $L_{1/2}$ Works Better

- ▶ Less bias in large coefficients
- ▶ Strong sparsity enforcement
- ▶ Faster convergence near solution

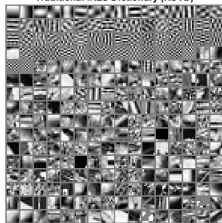
Method	Behavior
L_1	Smooth but biased
$L_{1/2}$	Promotes Sparsity

Dataset

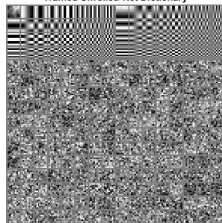
- ▶ BSDS500 dataset
- ▶ Experiments on:
 - ▶ Single image
 - ▶ 20 random images
- ▶ Same:
 - ▶ Dictionary
 - ▶ Noise level
 - ▶ Patch extraction

Dictionaries Learned

Traditional IRLS Dictionary (KSVD)



Trained Unrolled-Net Dictionary



Denoising Results

Noisy (RMSE: 0.0980)



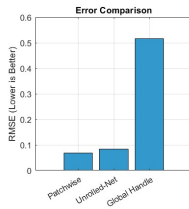
1. Patchwise
PSNR: 23.22 | RMSE: 0.0690



2. Unrolled-Net
PSNR: 21.50 | RMSE: 0.0842



3. Global
PSNR: 5.74 | RMSE: 0.5163



Quantitative Results

Method	PSNR	RMSE	Time
OMP	20.81	0.091	2.59
L_1	23.75	0.064	121.87
$L_{1/2}$	24.70	0.058	18.95

Unrolled IRLS-Net

- ▶ Idea: Convert iterative IRLS into a fixed network
- ▶ Each iteration $\rightarrow$ one layer in the network
- ▶ Pipeline:

Input $\rightarrow$ [Weight Update $\rightarrow$ Least Squares] $\times K \rightarrow$ Sparse Code

- ▶ Advantage:
 - ▶ Faster inference
 - ▶ Fixed computation time

Why Unrolling Helps

- ▶ Traditional IRLS:
 - ▶ Iterative and slow
- ▶ Unrolled IRLS-Net:
 - ▶ Fixed number of layers (e.g., 5)
 - ▶ No need for convergence checks
- ▶ Key insight:
 - ▶ Approximate optimization with a learned structure

IRLS Variants Comparison

Method	Mode	PSNR	RMSE	Time
IRLS	Distinct	24.70	0.0582	18.95
IRLS	Sliding	22.92	0.0715	70.88
Unrolled-Net	Distinct	22.65	0.0737	5.86
Unrolled-Net	Sliding	21.50	0.0842	22.48
Global Handle	Global	5.74	0.5163	651.01

- ▶ Distinct patches perform better than sliding
- ▶ Unrolled-Net is faster but slightly less accurate

Insights from IRLS Variants

- ▶ Sliding patches:
 - ▶ Reduce blocking artifacts
 - ▶ But significantly increase computation
- ▶ Unrolled-Net:
 - ▶ Faster (3x)
 - ▶ Slight drop in reconstruction quality
- ▶ Global approach:
 - ▶ Poor performance → not suitable

Observations

- ▶ OMP:
 - ▶ Fast but low quality
- ▶ L_1 :
 - ▶ Good quality but slow
- ▶ $L_{1/2}$:
 - ▶ Best balance
 - ▶ High PSNR + moderate time

Challenges Faced

- ▶ **Non-convex Optimization**
 - ▶ Sensitive to initialization
 - ▶ Can converge to local minima
- ▶ **Hyperparameter Tuning**
 - ▶ Different λ needed for different methods
 - ▶ Required extensive experimentation
- ▶ **Sliding Patch Trade-off**
 - ▶ Better visual quality
 - ▶ Much higher runtime

Conclusion

- ▶ $L_{1/2}$ outperforms OMP and L_1
- ▶ Provides better sparsity and reconstruction
- ▶ Practical and effective for denoising

Thank You!